

$$h \rightarrow 6.3 \quad \checkmark$$

$$h \rightarrow 6.4 \quad \checkmark$$

$$h \rightarrow 6.5 \quad \checkmark$$

$$h \rightarrow 3 \quad \checkmark$$

$$h \rightarrow 4 \quad \checkmark$$

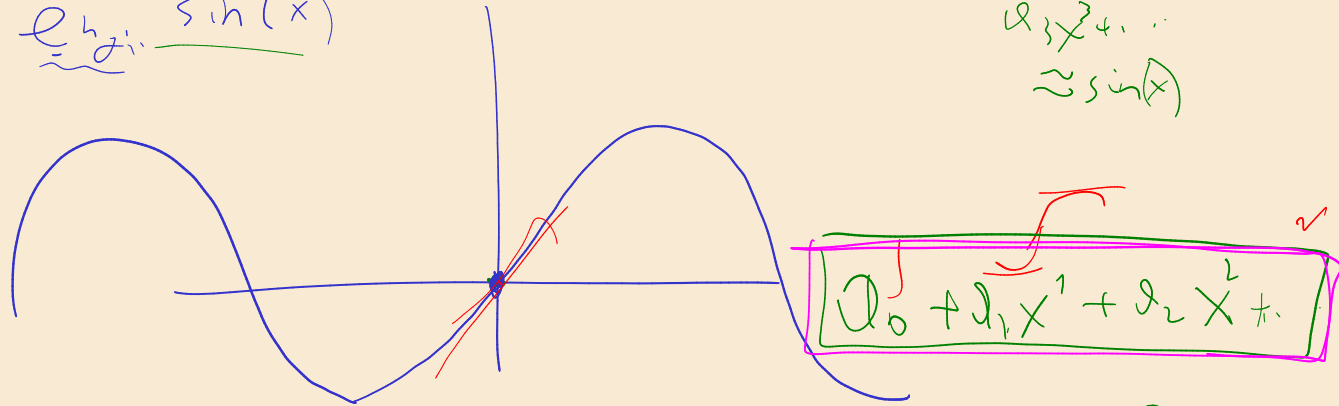
$$h \rightarrow 6 \quad \checkmark$$

$$h \rightarrow 6 \quad \checkmark$$

~~11~~

42

$$ML \quad e^{h x_i} \sin(x)$$



$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \approx \sin(x)$$

$$p(0) = \sin(0) = a_0 = 0$$

$$p'(0) = \sin'(0) = a_1$$

$$p''(0) = \frac{\sin''(0)}{2!} = a_2 \rightarrow \frac{\sin^{(n)}(0)}{n!} = a_n$$

$$1 a_1 + 2 a_2 x + 3 a_3 x^2 + \dots$$

$$2 a_2 + 3 \cdot 2 a_3 x + \dots x^2 \dots$$

$$2! a_2 = \sin''(0)$$

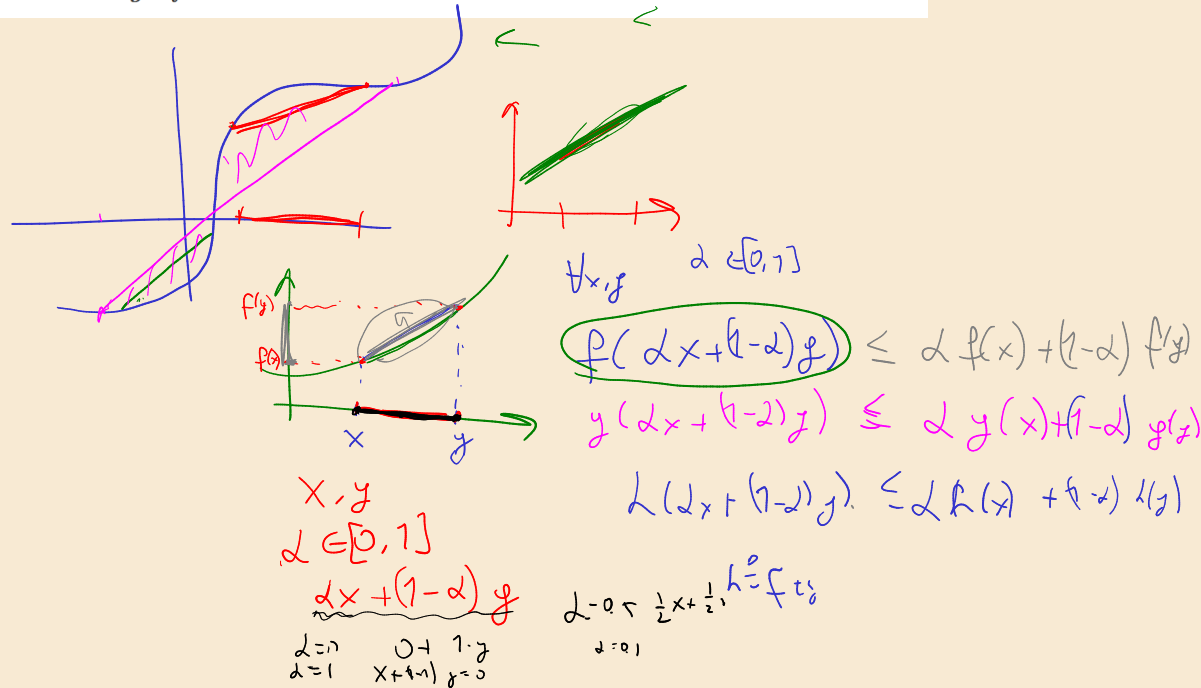
$$\frac{f^{(k)}(a)}{k!} (x-a)^k \rightarrow \sin x$$

2, 3

### 03 Convex Function Properties

Consider two convex functions  $f, g : \mathbb{R} \rightarrow \mathbb{R}$ .

1. Show that  $f + g$  is convex.
2. Now, assume that  $g$  is additionally non-decreasing, i.e.,  $g(y) \geq g(x)$  for all  $x \in \mathbb{R}$ , for all  $y \in \mathbb{R}$  with  $y > x$ . Show that  $g \circ f$  is convex.



$$f, g \text{ convex}$$

$$g(y) \geq g(x)$$

$$y > x$$

$$g \circ f$$

$$h(x) = g(f(x))$$

$$g(f(\alpha x + (1-\alpha)y)) \leq g(\alpha f(x) + (1-\alpha)f(y))$$

$$g(\alpha f(x) + (1-\alpha)f(y)) \leq \alpha g(f(x)) + (1-\alpha)g(f(y))$$

$$h(\alpha x + (1-\alpha)y) \leq \alpha h(x) + (1-\alpha)h(y)$$

## 🏠 🏠 04 Testing Convexity in ML Functions

Determine whether the following ML-related functions are convex, concave, or neither on the given intervals:

1. **Mean Squared Error:**  $L(w) = \frac{1}{2}(w - 3)^2$  on  $\mathbb{R}$
2. **ReLU Activation:**  $\text{ReLU}(x) = \max(0, x)$  on  $\mathbb{R}$
3. **Sigmoid Function:**  $\sigma(x) = \frac{1}{1+e^{-x}}$  on  $\mathbb{R}$

$$(w-3)(w-3)'$$

1.  2.  $\frac{1}{2}(w-3) \cdot 1$   $f''(w) \geq 0$

Consider the logistic loss function:

$$\ell(z; y) = -y \log \sigma(z) - (1 - y) \log(1 - \sigma(z))$$

where  $\sigma(z) = \frac{1}{1+e^{-z}}$  is the sigmoid function,  $z$  is the logit (linear combination of features), and  $y \in \{0, 1\}$  is the true binary label.

1. **Task:** Show that  $\frac{d}{dz} \ell(z; y) = \sigma(z) - y$ .

2. **Bonus:** Check if the function  $g(z) = (y - \sigma(z))^2$  is convex with respect to  $z$ .

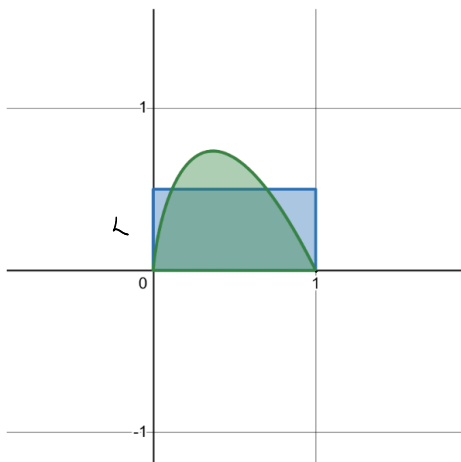
$$\begin{aligned} 0, & x < 0 \\ 1, & x > 0 \\ \emptyset, & x = 0 \end{aligned}$$

$$\begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0.9 - 0.9 \\ 1 - 0.1 \end{pmatrix}^2$$

$$(f(x) - y)^2$$

### 03 Equal Areas (textbook 6.3) [↗](#)

Below is the graph of  $f(x) = -x \ln(x^2)$  on  $[0, 1]$ :



Green region (under the curve) and blue rectangle (over  $[0, 1]$ ) have equal areas; find the rectangle's height.

The green region (under the curve, above the  $x$ -axis) and the blue rectangle (over  $[0, 1]$ ) have the same area. What is the height of the blue rectangle?

$$\int_0^1 x \ln(x^2) dx =$$

$$= \int_0^1 -x 2 \ln x dx =$$

$$= -2 \int_0^1 x \ln x dx =$$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$dv = x dx \rightarrow v = \frac{x^2}{2}$$

$$\int u dv = uv \Big|_0^1 - \int v du$$

$$u \rightarrow du$$

$$dv \rightarrow v$$

$$-2 \left( \ln x \cdot \frac{x^2}{2} \right) \Big|_0^1 - \int_0^1 \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$- \ln x x^2 \Big|_0^1 + \int_0^1 x dx =$$

$$= - \left( 1 \ln 1 + \frac{1^2}{2} \right) \rightarrow \frac{1}{2}$$

## 04 Convolution (textbook 6.4)

There are several ways to combine two functions into a new one. One that plays a central role in machine learning is called **convolution**.

The convolution of two functions  $f$  and  $g$ , denoted  $f * g$ , is defined by:

$$(f * g)(x) = \int_{-1}^1 f(y) g(x - y) dy$$

(Here  $x$  is fixed; the integration variable is  $y$ .)

Compute  $(f * g)(0)$  for  $f(x) = x^2$  and

$$g(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

Hint: Plug in  $g$ 's formula, then split  $[-1, 1]$  into the regions where  $g(x - y) = 0$  and where  $g(x - y) = 1$ .



$$x=0 \quad f * g = \int_{-1}^1 f(y) g(y) dy = \int_{-1}^1 y^2 g(-y) dy = \int_{-1}^0 y^2 \cdot 1 dy + \int_0^1 y^2 \cdot 0 dy$$

$$g(-y) = 1 \text{ for } -y > 0 \Rightarrow y < 0$$

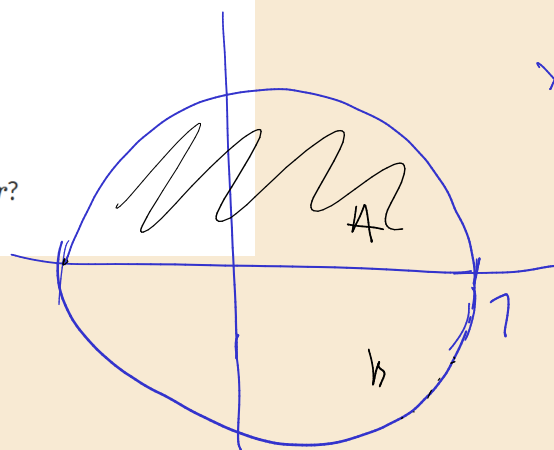
$$g(-y) = 0 \text{ for } -y \leq 0 \Rightarrow y \geq 0$$

$$\int_{-1}^0 y^2 dy = \left. \frac{y^3}{3} \right|_{-1}^0 = \frac{1}{3} + 0$$

$$\int_a^b f = F(x) \Big|_a^b = F(b) - F(a)$$

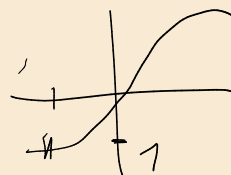
Derive the area formula of a circle of radius  $r$  from first principles. Start with the unit circle  $x^2 + y^2 = 1$ .

- Express  $y$  in terms of  $x$ ; over what interval does  $x$  vary?
- Which function  $f(x)$ 's integral do you need?
- Use the substitution  $x = \sin t$ . What interval does  $t$  vary over?
- What expression appears under the integral sign?
- Use the identity  $\cos^2 t = \frac{1 + \cos(2t)}{2}$  to evaluate it.
- What is the area of the unit circle?
- What linear transformation do you apply to get a circle of radius  $r$ ?
- By what factor does the area scale?



$$A = \pi$$

$$2 \cdot A$$



$$\int_{-\pi/2}^{\pi/2}$$

$$x^2 + y^2 = 1$$

$$y = \pm \sqrt{1 - x^2}$$

$$x \in [-1, 1]$$

$$y = \sqrt{1 - x^2}$$

$$\int \sqrt{1 - x^2} dx$$

$$(-1)$$

$$x = \sin t$$

$$dx = \cos t dt$$

$$\cos t \cdot \cos t dt =$$

$$\int_{-\pi/2}^{\pi/2} \underbrace{\sqrt{1 - \sin^2 t}}_{\cos^2 t} dt$$

$$\int_{-\pi/2}^{\pi/2} \cos^2 t dt$$

$$\cos^2 t = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \cos(2t)}{2} =$$

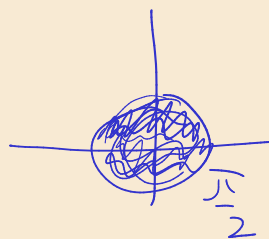
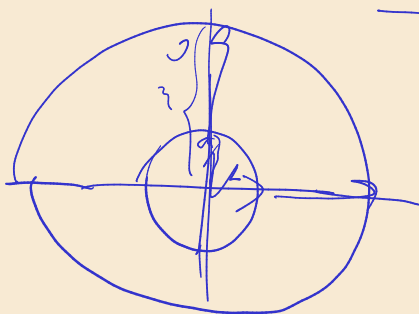
$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos 2t =$$

$$= \left[ \frac{1}{2} t + \frac{\sin(2t)}{4} \right] \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} =$$



$$= \frac{\pi}{4} - \frac{\pi}{4} = \frac{\pi}{2}$$

$\cos^2 t$



$$\odot \rightarrow 2 \frac{\pi}{2} \rightarrow \pi$$

$$2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\pi \rightarrow \pi \det \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \pi \cdot 2^2$$